

Tutorial 10 Nonlinear Bayesian Regression 10.12

Gaussian Process Regression

Setup:

input $x \in \mathbb{R}^d$, param: $\theta \in \mathbb{R}^P$, transform: $\phi(x)$

$$f(x, \theta) = \phi(x)^T \theta$$

noisy output: $y = f(x, \theta) + \epsilon$ $\epsilon \sim \mathcal{N}(0, \sigma^2)$

prior: $\theta \sim \mathcal{N}(0, P)$

$$D = \{(x_i, y_i)\}_{i=1}^N$$

posterior: $p(\theta | D) = \mathcal{N}(\theta | \hat{\mu}_0, \hat{\Sigma}_0)$

$$\hat{\mu}_0 = (P^{-1} + \Phi^T \Phi)^{-1} \Phi^T y$$

$$\hat{\Sigma}_0 = (P^{-1} + \Phi^T \Phi)^{-1}$$

$$\Phi = [\phi(x_1), \phi(x_2), \dots, \phi(x_N)] \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

predictive: given x_*

$$p(f_* | x_*, D) = \mathcal{N}(f_* | \hat{\mu}_*, \hat{\sigma}_*^2)$$

$$\begin{cases} \hat{\mu}_* = \phi(x_*)^T \hat{\mu}_0 \\ \hat{\sigma}_*^2 = \phi(x_*)^T \hat{\Sigma}_0 \phi(x_*) \end{cases}$$

a) predictive mean:

$$\hat{\mu}_* = \phi_*^T \hat{\mu}_0 = \phi_*^T (P^{-1} + \underbrace{\Phi^T \Phi}_{B^T R^T})^{-1} \underbrace{\Phi^T}_{B^T} \underbrace{y}_{R^T}$$

ps 1-15

$$(P^{-1} + B^T R^T)^{-1} B^T R^T = P B^T (B P B^T + R)^{-1}$$

$$\hat{\mu}_* = \phi_*^T P \Phi (\Phi^T P \Phi + \sigma^2 I)^{-1} y$$

b) predictive variance

$$\hat{\sigma}_*^2 = \phi_*^T \hat{\Sigma}_0 \phi_* = \phi_*^T (P^{-1} + \Phi^T \Phi)^{-1} \phi_*$$

ps 1-15

$$(A^{-1} + U C^{-1} U^T)^{-1} = A - A U (C + U^T A U)^{-1} U^T A$$

$$\hat{\sigma}_*^2 = \phi_*^T (P - P \Phi (\sigma^2 I + \Phi^T P \Phi)^{-1} \Phi^T P) \phi_*$$

$$= \phi_*^T P \phi_* - \phi_*^T P \Phi (\sigma^2 I + \Phi^T P \Phi)^{-1} \Phi^T P \phi_*$$

c) let $k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$ prior on θ is embedded into the kernel

$$\Phi^T P \Phi = [\phi(x_1), \dots, \phi(x_N)]^T P [\phi(x_1), \dots, \phi(x_N)] = \underline{K}$$

$$= [k(x_i, x_j)]_{ij} \quad \text{kernel matrix}$$

$$\Phi^T P \phi_* = [\phi(x_1), \phi(x_2), \dots, \phi(x_N)]^T P \phi_* = \underline{K}_* = [k(x_i, x_*)]_i$$

$$\phi_*^T P \phi_* = k_{**} = k(x_*, x_*) \quad \text{kernel test vector}$$

$$\Rightarrow \hat{\mu}_* = K_*^T (K + \sigma^2 I)^{-1} y, \quad \hat{\sigma}_*^2 = k_{**} - K_*^T (K + \sigma^2 I)^{-1} K_*$$

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predictive mean:

$$\hat{\mu}_* = \underbrace{K_X^T (K + \sigma^2 I)^{-1}}_{\text{has } x^*} y \quad \Rightarrow Z = (K + \sigma^2 I)^{-1} y$$

depends on training data

$$\hat{\mu}_X = K_X^T Z = \sum_{i=1}^n K(x_i, x_*) \cdot z_i$$

linear kernel: $K(x_i, x_j) = \alpha x_i^T x_j \quad \alpha > 0$

$$\hat{\mu}_* = \sum_{i=1}^n \alpha x_i^T x_* z_i = \underbrace{\left(\sum_{i=1}^n \alpha x_i^T z_i \right)}_{\beta^T} \cdot x_* = \underbrace{\beta^T x_*}_{\text{linear}}$$

polynomial kernel: $K(x_i, x_j) = \alpha (x_i^T x_j + 1)^2$

$$\Rightarrow \hat{\mu}_* = \sum_{i=1}^n \alpha (x_i^T x_* + 1)^2 z_i = \sum_{i=1}^n \alpha (\underbrace{x_i^T x_*}_{A} + 1) z_i$$

$$= x_*^T \underbrace{\left(\sum_{i=1}^n \alpha x_i x_i^T \right)}_A x_* + \underbrace{\left(2 \sum_{i=1}^n \alpha z_i x_i^T \right)}_B x_* + \underbrace{\sum_{i=1}^n \alpha z_i}_C$$

= $x_*^T A x_* + B x_* + C$: quadratic func of x_*

RBF kernel: $K(x_i, x_j) = \alpha e^{-\alpha \|x_i - x_j\|^2}$

$$\hat{\mu}_* = \sum_{i=1}^n \underbrace{\alpha z_i}_{\text{weight (pos, neg)}} e^{-\alpha \|x_i - x_*\|^2}$$

mean

